

The Visualisation Design Process

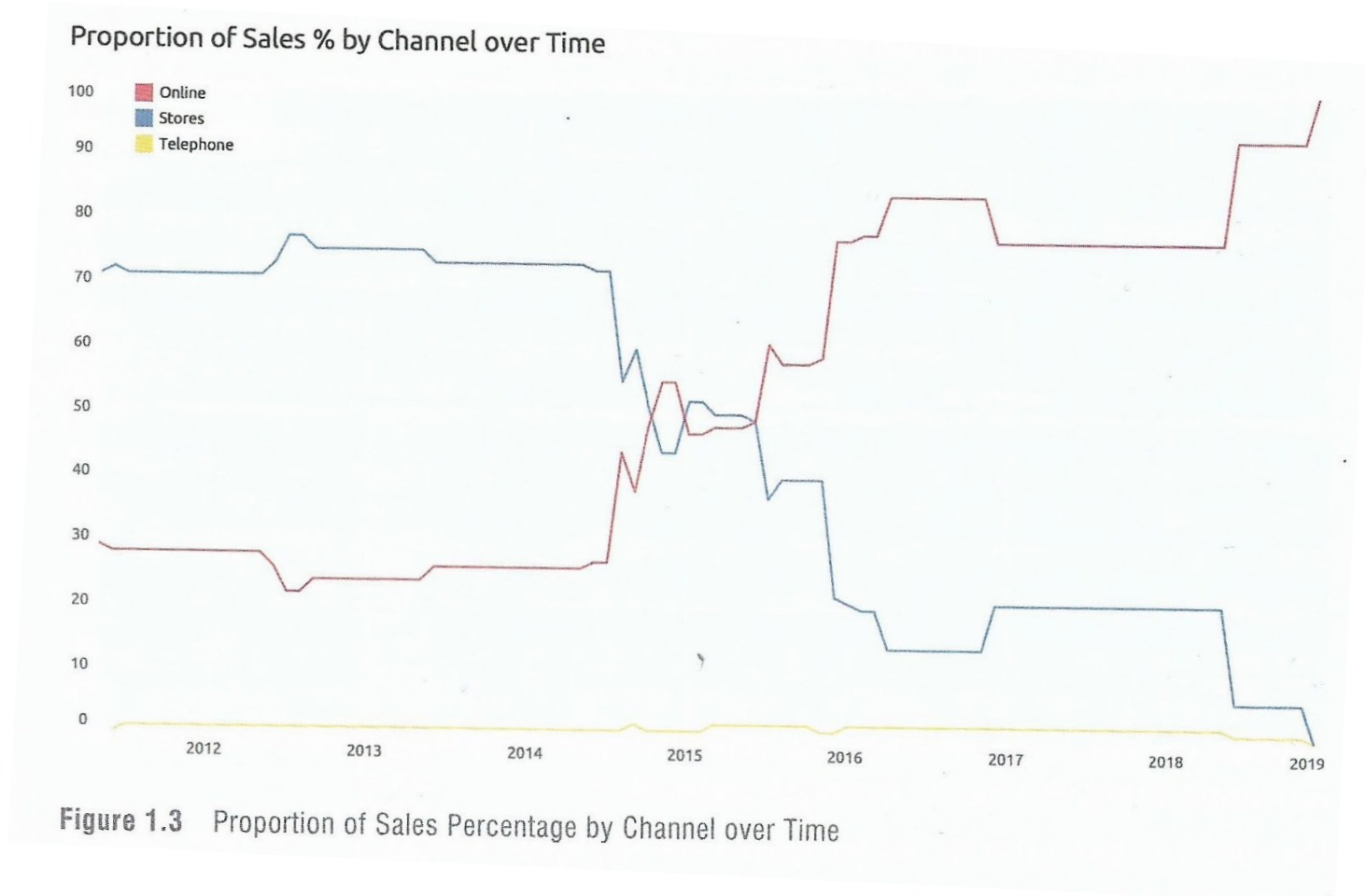
CM4125 – Topic 2

Based on Chapter 1-3 of Andy Kirk's "Data Visualisation" book

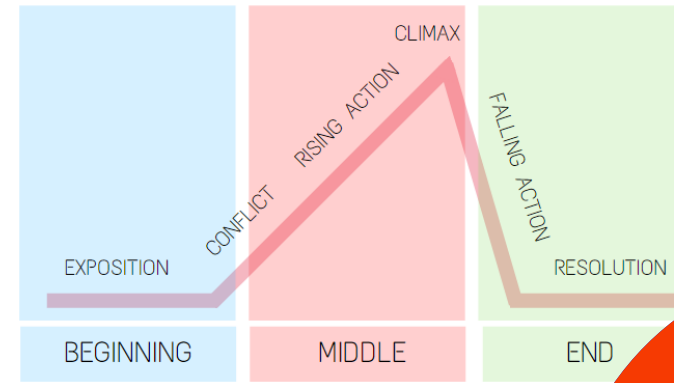
Data Vis Foundations

Basic Definitions

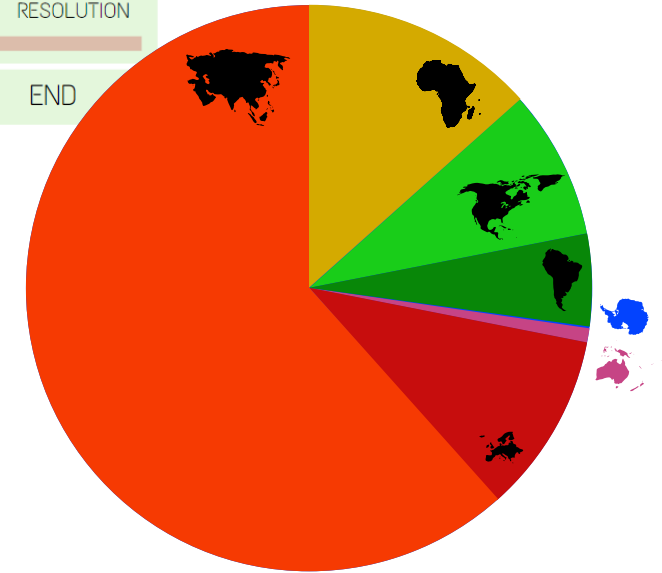
- Data viz encompasses the representation and presentation of data to facilitate understanding
- Its building blocks are marks and attributes:
 - Marks: Points, lines or shapes used to represent the items of data
 - Attributes: Visual variations of the marks (scales, sizes, colours, etc.)



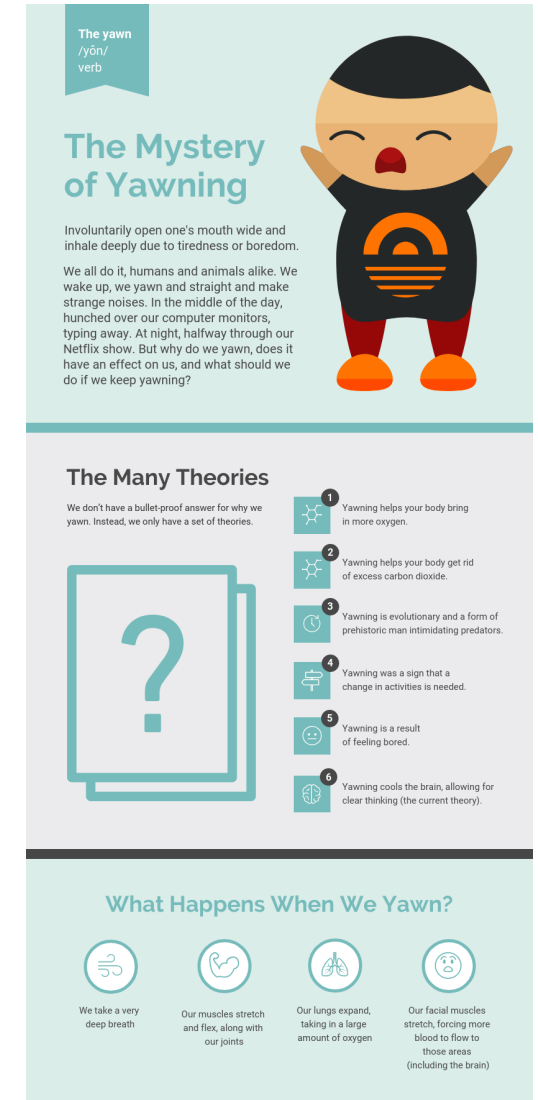
- Number story: Any type of visual representation that informs an audience based on statistical data
 - A plot
 - A phrase
 - An image
 - etc.



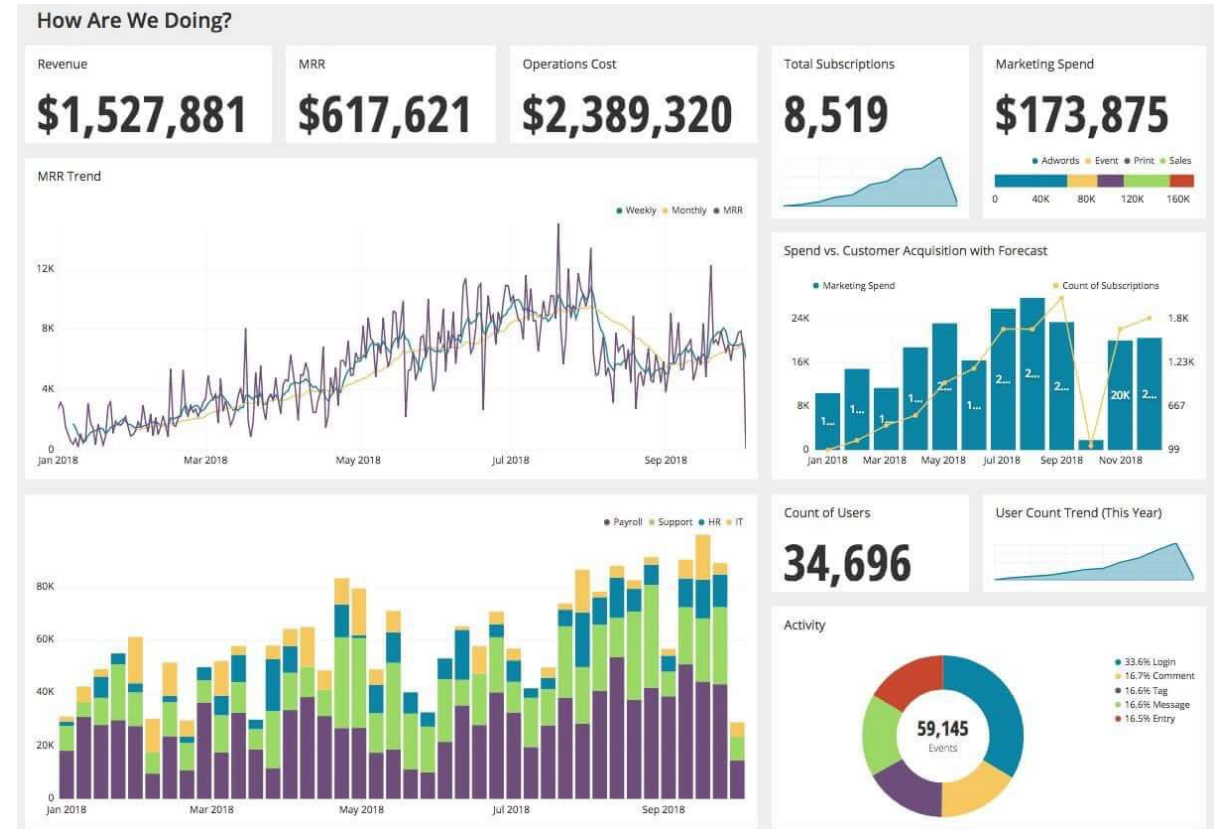
DID YOU KNOW?
Every 35 days, your skin replaces itself and your body makes new cells based on the foods you eat.



- Infographic:
 - A collection of images, charts and “minimal” text that gives an easy-to-read overview of a topic.
 - A visual representation of information or data (Oxford Dict.)
- It is a STILL representation, so data and visualisations must be chosen carefully.
- There are many online tools that help you design them e.g. Piktochart, Infograpia, etc.

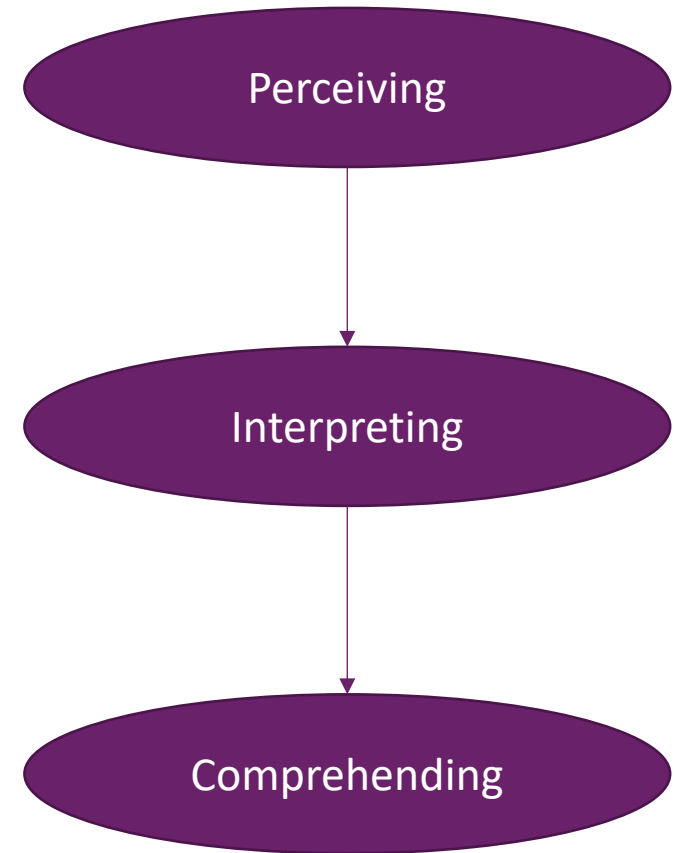


- Dashboard:
 - An information management tool used to track, analyse and display **key performance indicators (KPIs)**, metrics and data points related to a specific issue
- Customisable reports
- Dynamic and with minimal text
- Require more idea of what the user may need to see/understand
- Still, many tools to do them! (e.g. Python, Tableau, PowerBi, etc.)

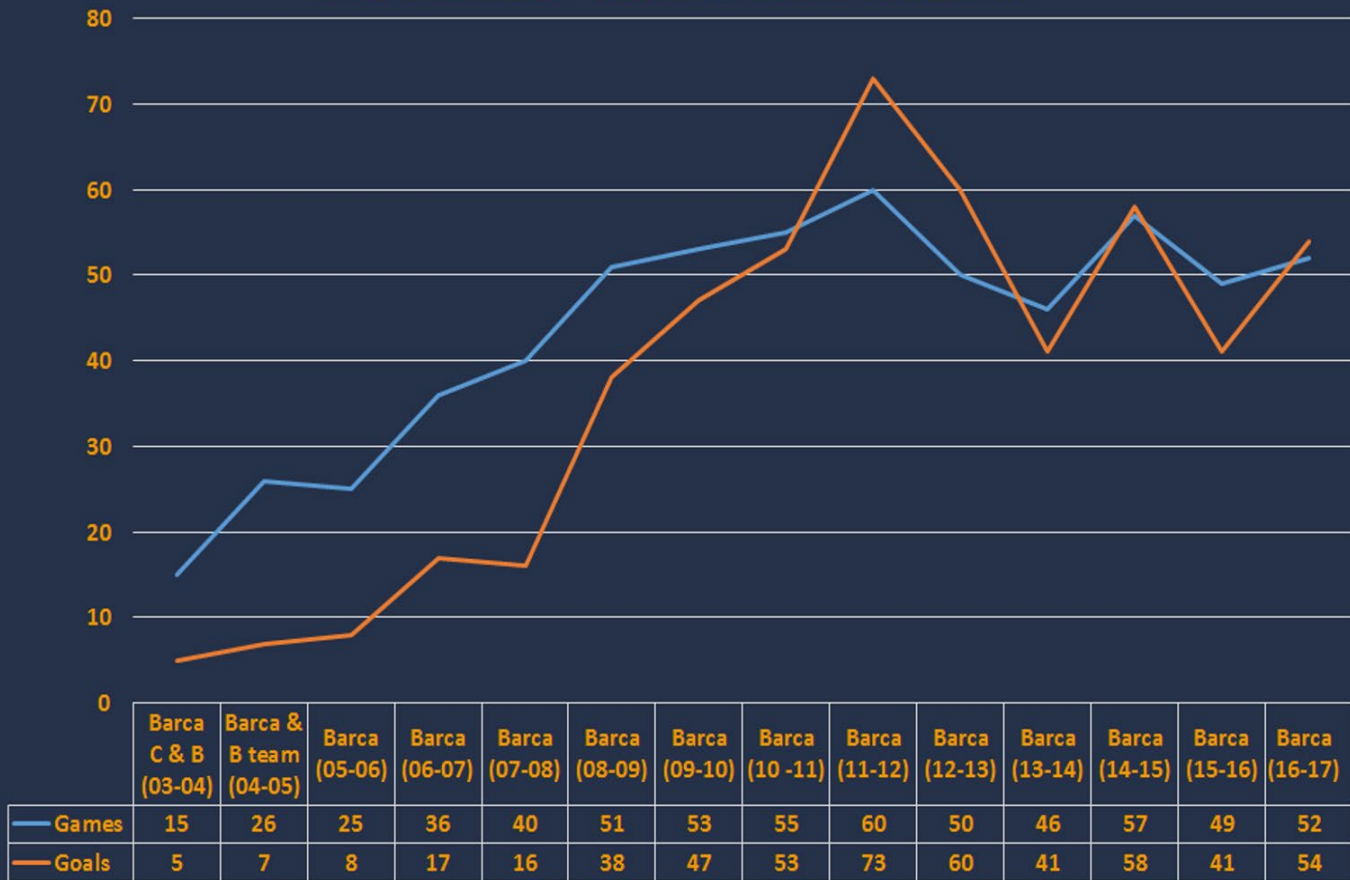


Presentation

- How we choose to package the visualisation
 - Interactivity
 - Features or annotations
 - Colour usage
 - Composition of the work
- Connection between presentation and representation!
- Facilitate understanding



Lionel Messi At 30 - Season-By-Season Club Record



Total sightings of Winglets and Sprungles



Coming back to the first example...

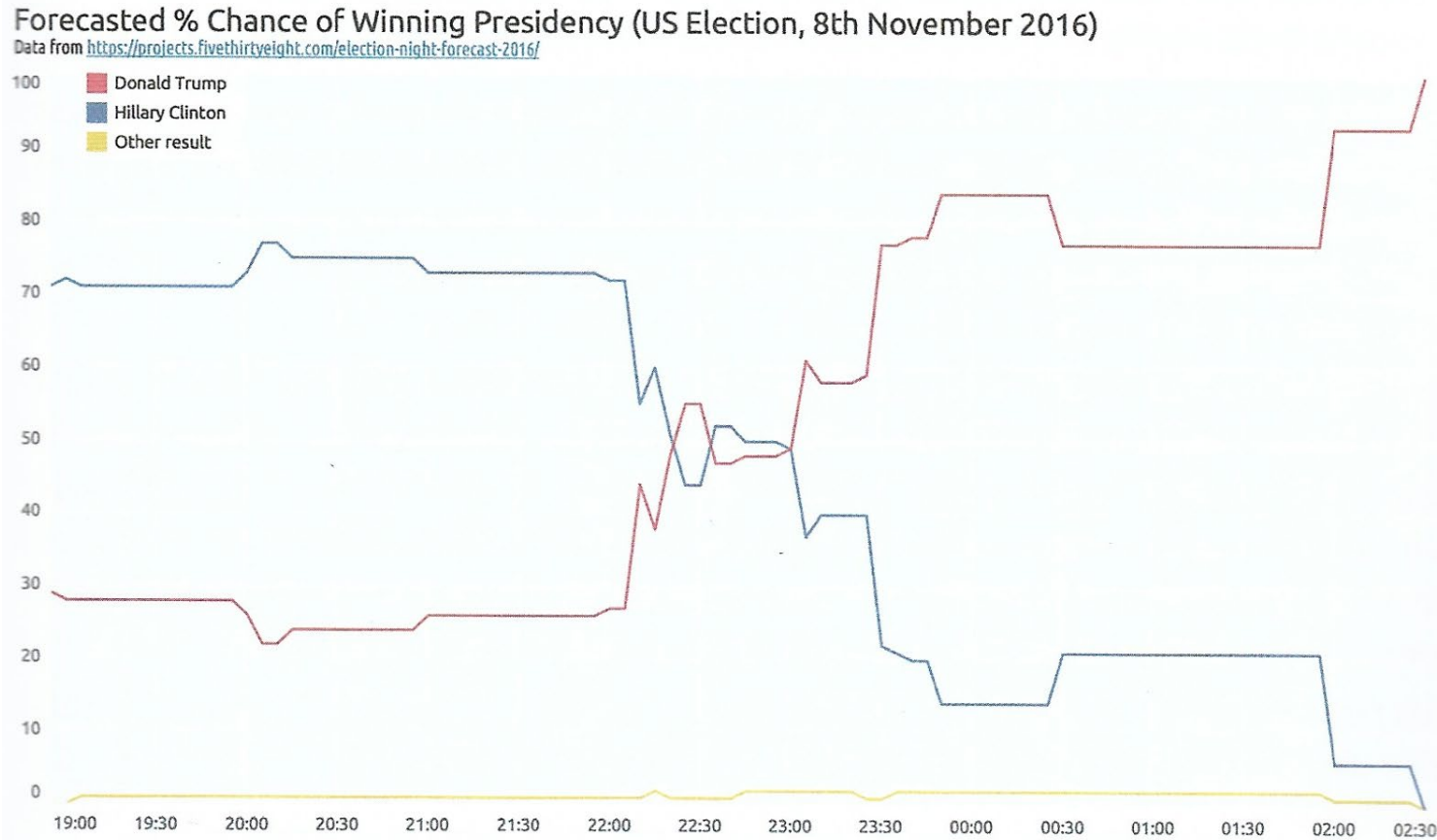
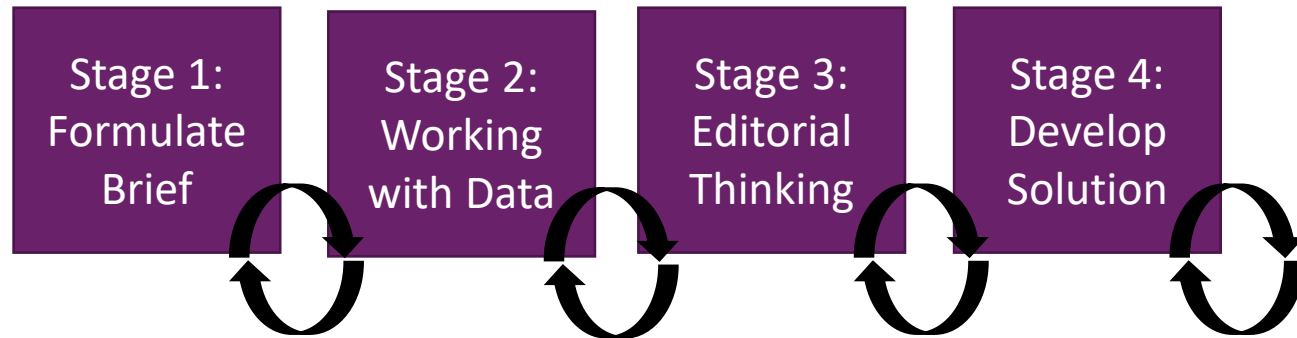


Figure 1.7 Forecasted % Chance of Winning Presidency (US Election, 8 November 2016)

The perfect viz doesn't exist!

- It is an iterative process
- It may never please everyone
- Effective decisions efficiently made



Some observations (the 12 steps?)

1. Reducing the randomness of your approach
2. Every project is different
3. Adaptability
4. Protect experimentation
5. The first occasion, not the last (unless you are doing your coursework at 11 pm one day before the deadline)
6. Time management.
7. Mindsets
8. Documenting
9. Communication
10. Attention to detail
11. Kill your darlings
12. Learn

The 3 Principles of Design

- Good visualisation is **trustworthy**
 - Is it reliable?
- Good visualisation is **accessible**
 - Is it usable?
- Good visualisation is **elegant**
 - Is it aesthetic?

**Put in other words by Dieter Rams
(German designer)**

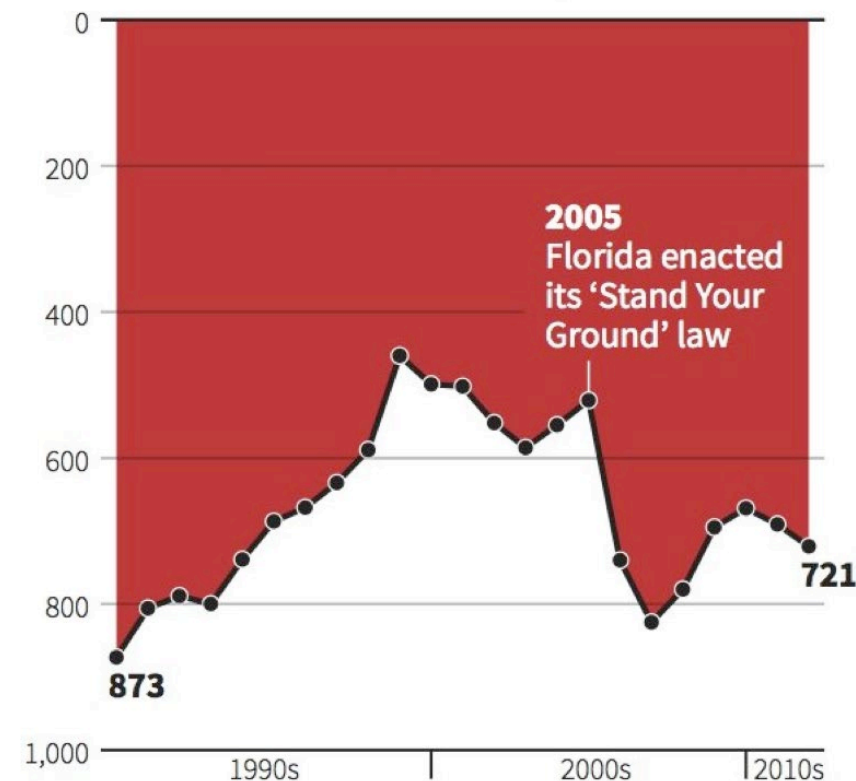
- innovative
- useful
- aesthetic
- understandable
- unobtrusive
- honest
- long-lasting
- thorough to the last detail
- environmentally friendly
- AS LITTLE DESIGN AS POSSIBLE!

1. Trustworthy

- Trust $\neq$ truth
 - Achieving trust is an aim, presenting truth is an obligation!
 - *“Trust arrives on foot and goes away on horseback.” (Dutch proverb)*
- Mostly, this can be achieved by avoiding “bullsh*t” (as we will see)

Gun deaths in Florida

Number of murders committed using firearms

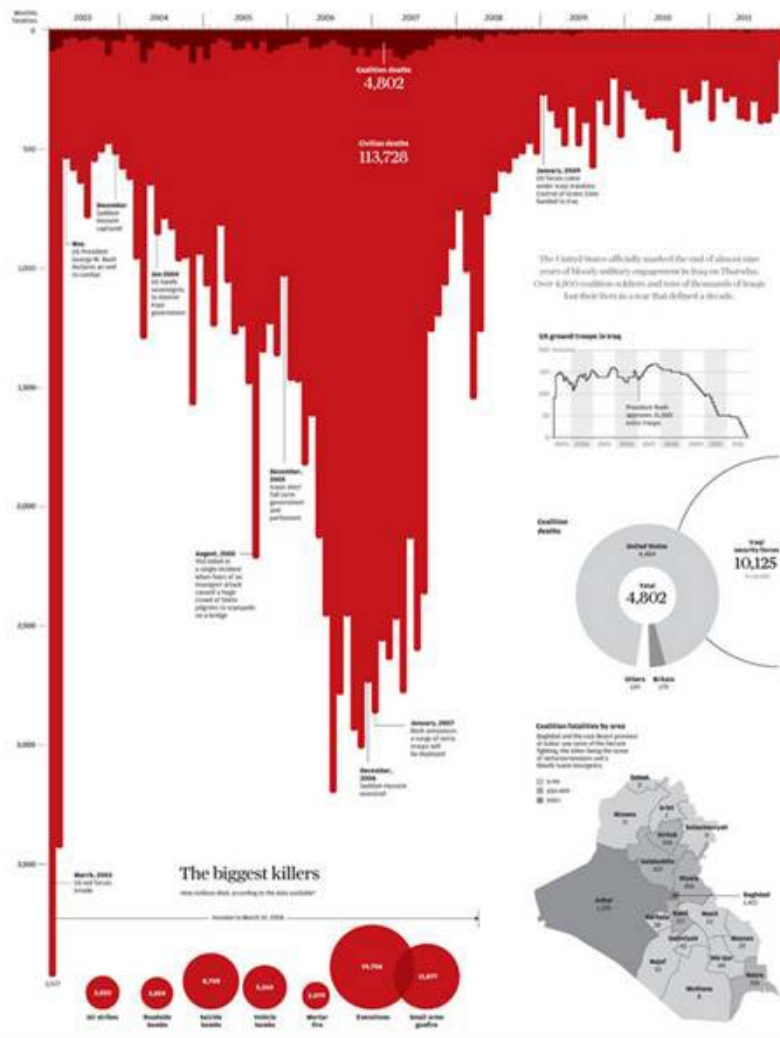


Source: Florida Department of Law Enforcement

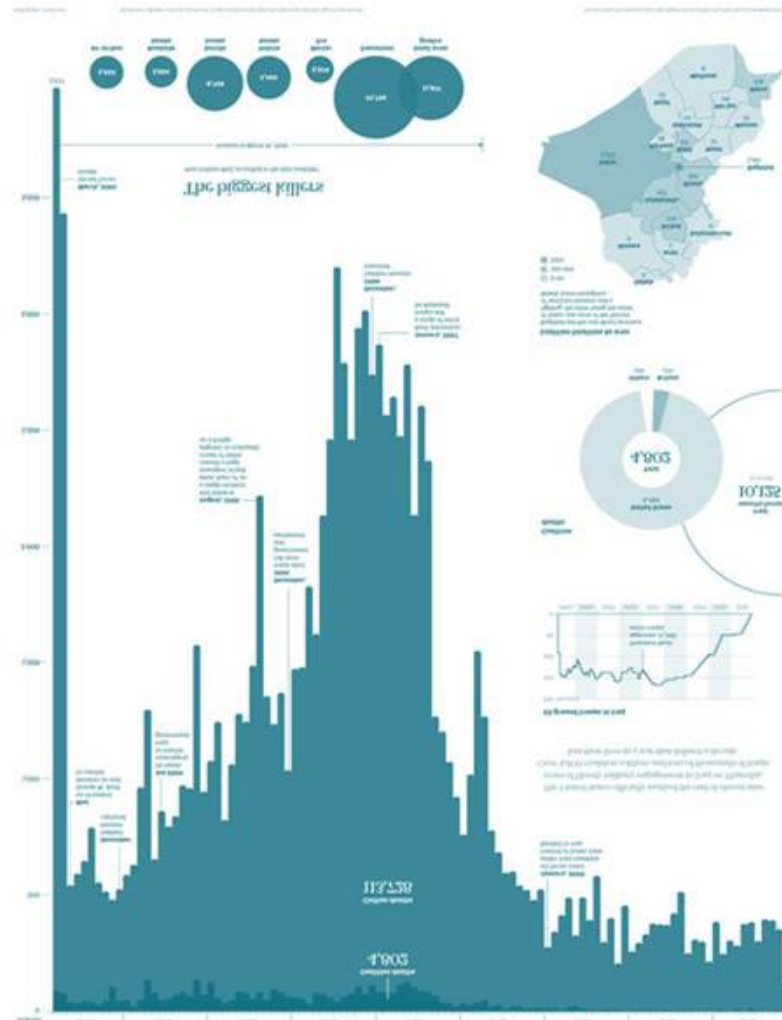
C. Chan 16/02/2014

REUTERS

Iraq's bloody toll



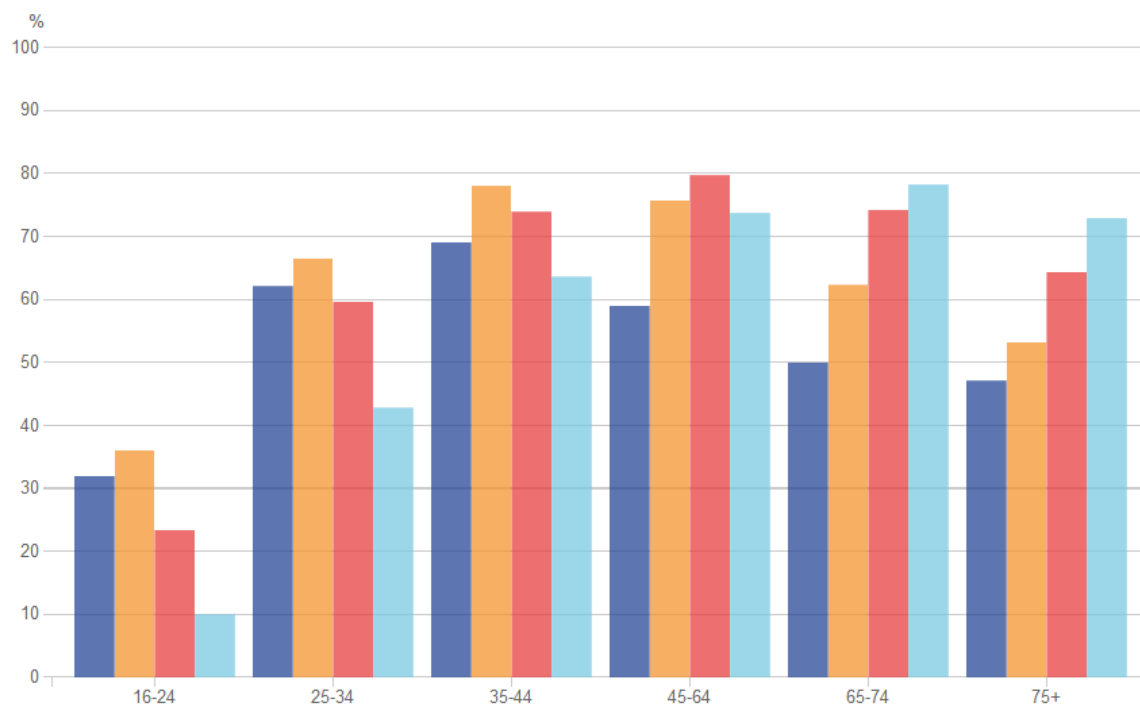
Iraq: Deaths on the decline



Example

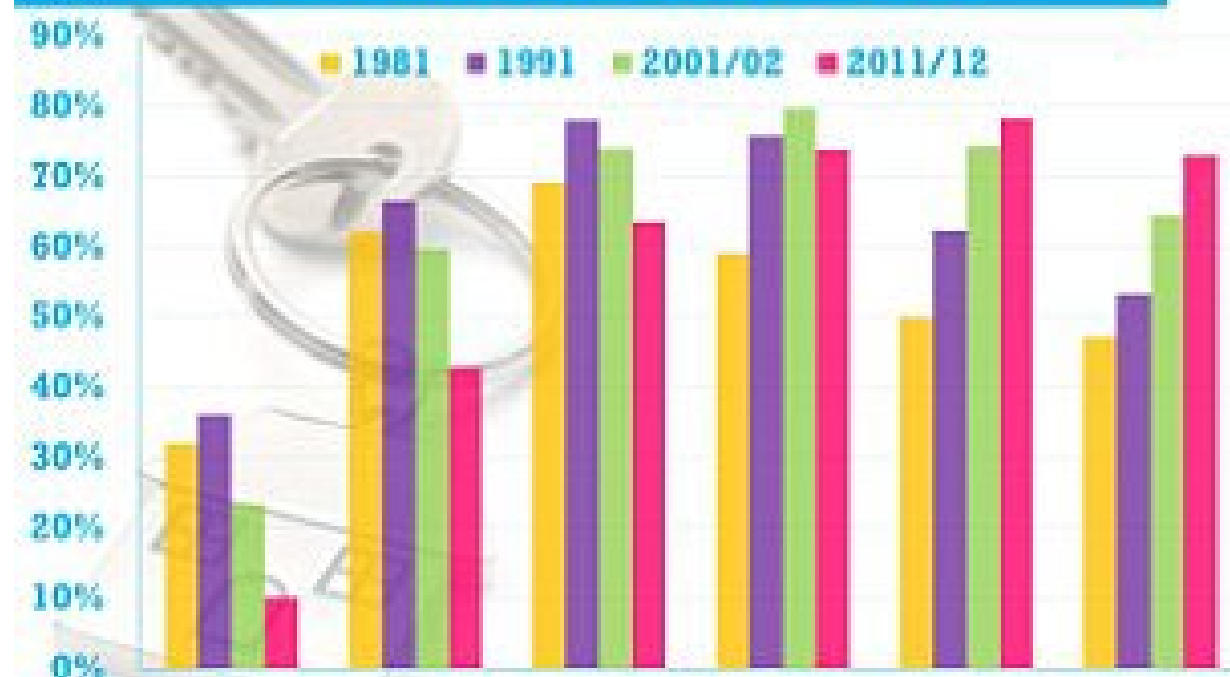
Percentage of each age group that are home owners⁵, England, 1981 to 2012

Calendar years Financial years
■ 1981 ■ 1991 ■ 2001/02 ■ 2011/12



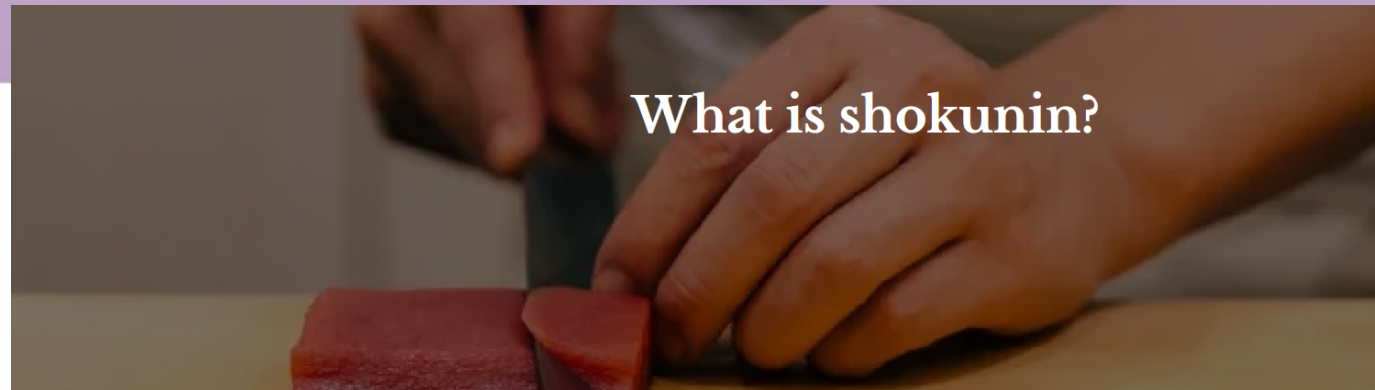
Source: English Housing Survey (EHS) 2012 to 2013, Table FC2101, DCLG ; EHS 2001/02, Table S106, DCLG

FALLING NUMBER OF YOUNG HOMEOWNERS

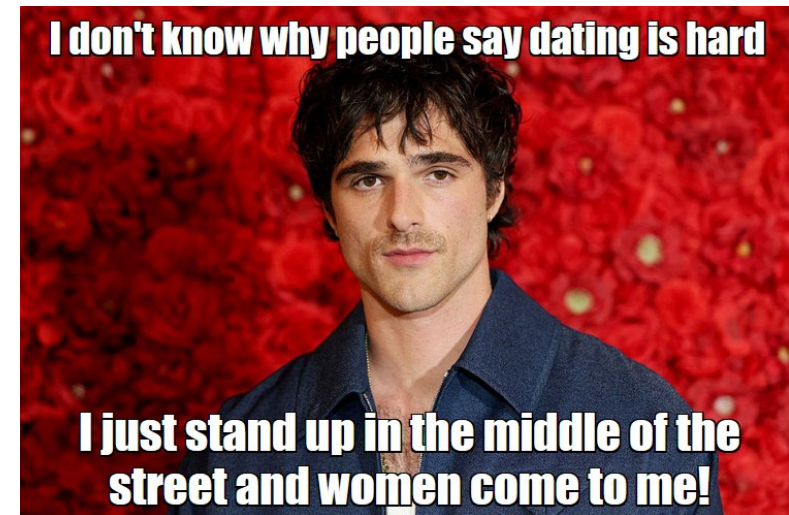


2. Accessible

- Understanding the subject
 - Simple $\neq$ Easy
 - Complicated $\neq$ Complex
- Changing a diaper is simple (few steps), but not always easy!
- Landing on the moon is complicated (lots of steps), and managing a relationship is complex.
- CLARIFY (i.e. **try** to make it EASY) when the audience doesn't have the knowledge (or capacity) to handle a COMPLEX subject!
- SIMPLIFY when the audience has that knowledge and can handle a COMPLICATED subject but just don't know how you did it!



Shokunin (職人) is a Japanese term that embodies the essence of a skilled artisan or craftsman, and it goes beyond a mere job description. Shokunin represents a cultural and philosophical concept deeply ingrained in Japanese society, reflecting a commitment to mastery, craftsmanship and a profound connection to one's work.



2. Accessible

- Relevance
 - You may not have the answer, but you may be able to guide the audience into it
- Suitability
 - Minimum friction (with the audience)



About the Beamer class in presentation making

A short story

A. B. Arthur¹ J. Doe²

¹Faculty of Physics
Very Famous University

²Faculty of Chemistry
Very Famous University

Very Large Conference, April 2021

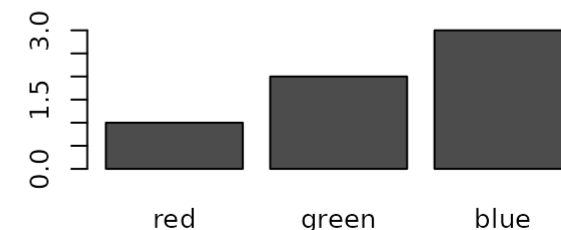
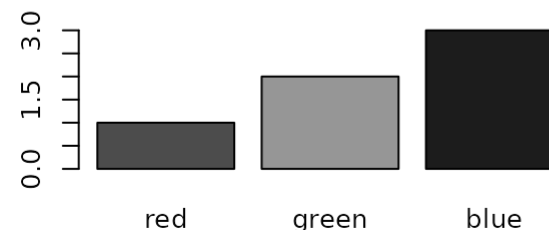
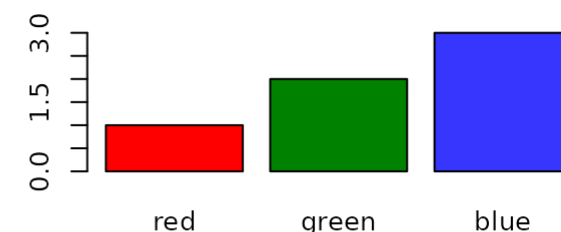
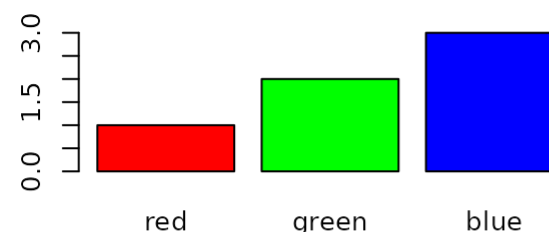
2.a. Accessible... but to whom?

- About **3 million people in the UK** (4.5% of the population) are colour blind.
 - **Two or three people** in a class of 50.
- It is not only colour vision: low vision, a phone screen in daylight, a washed-out projector, a printed handout, someone using a screen reader, etc.
- Any viz published by RGU (public body) falls under the **Public Sector Bodies Accessibility Regulations 2018**.
 - Our design preferences should be a legal duty!

Four considerations that cost you nothing

1. Colour must never be the only carrier ([WCAG 1.4.1](#))

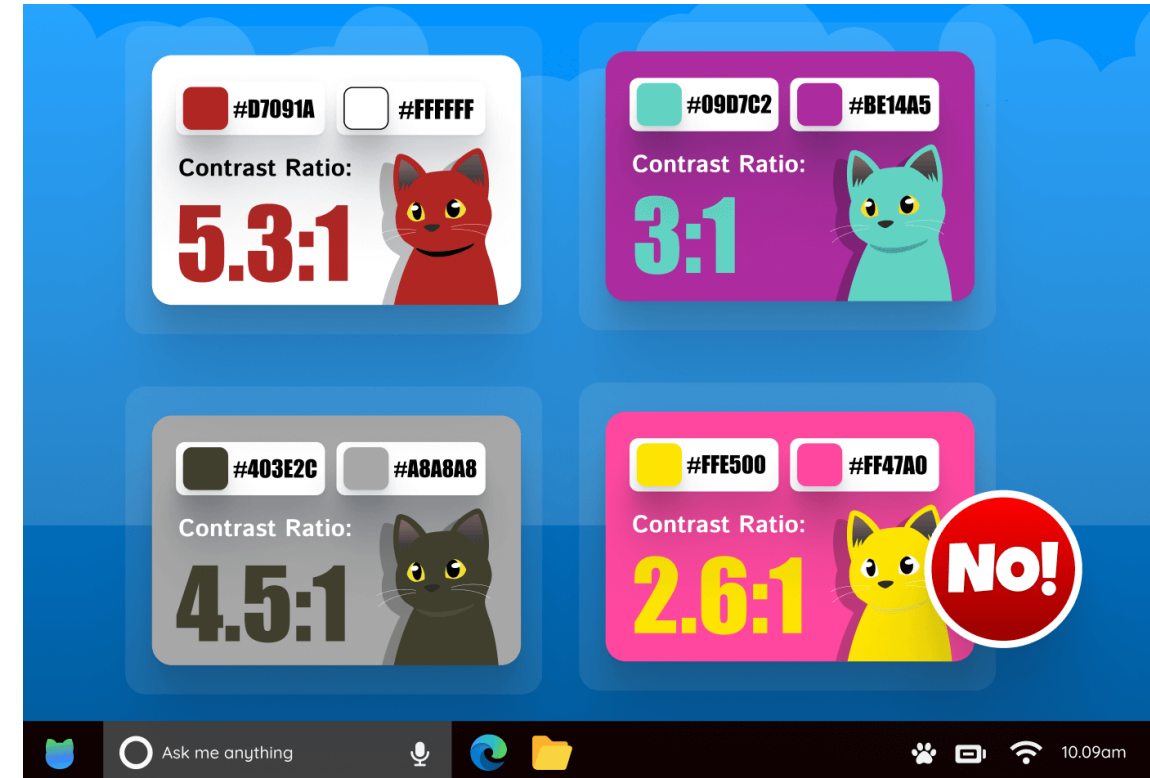
- If removing the colour destroys the message, the chart was never finished.
- Add a second, redundant encoding: **shape, dash pattern, line weight, direct labels, position, or plain text.**
- *How to test:* print it in greyscale!
"Get it right in black and white."



Four considerations that cost you nothing

2. Contrast is a number, not a subjective option!

- 4.5:1 for normal text ([WCAG 1.4.3](#)).
- 3:1 for the *graphical* parts ([WCAG 1.4.11](#)).
 - 1.4.11 asks each element to contrast with its **background**, not with other elements in series (for instance, two lines). Distinguishing series from *each other* is what rule 1 is for!



How to calculate the ratio?

- To do this calculation, you have to convert hex to relative luminance (RL) by

1. Split hex into RGB channels, e.g. #3A86FF would be

- **Red (R): 3A, Green (G): 86, Blue (B): FF**

2. Convert them into 0-255 integer values

- **Red (R): 3A → 58, Green (G): 86 → 134, Blue (B): FF → 255**

3. Normalise ($\div 255$) and gamma-linearise

- $LinearValue = 255 \times \left(\frac{Gamma\ Value}{255} \right)^{2.2}$

4. Apply the following formula to adjust the values:

- $L = (0.2126 \times R) + (0.7152 \times G) + (0.0722 \times B)$

5. Compute the contrast ratio:

- $Contrast\ Ratio = \frac{L_1 + 0.05}{L_2 + 0.05}$

For reference, **Maximum Contrast:** Pure White (#FFFFFF, L=1) vs Pure Black (#000000, L=0) gives $\{1 + 0.05\} / \{0 + 0.05\} = \mathbf{21:1}$.

OR USE THIS WEBSITE!

<https://webaim.org/resources/contrastchecker/>

Four considerations that cost you nothing

3. Use a palette that someone already designed

- **Okabe-Ito:** Eight categorical colours designed for colour-vision deficiency: in Python just paste the hex list; in Tableau, Edit Colors → Custom palette; in R, `palette.colors(palette = "Okabe-Ito")`
- **Continuous data → viridis** (matplotlib default, `scale_fill_viridis_c()` in ggplot, available in Tableau/Power BI as a custom ramp). Perceptually uniform *and* monotonic in lightness, so it translates well to greyscale.
- **Blue** looks roughly the same under every deficiency. If you need two colours, use **blue** and **orange**.

Okabe-Ito: 8 colours safe for colour-vision deficiency



viridis — for continuous data: perceptually uniform and monotonic in lightness



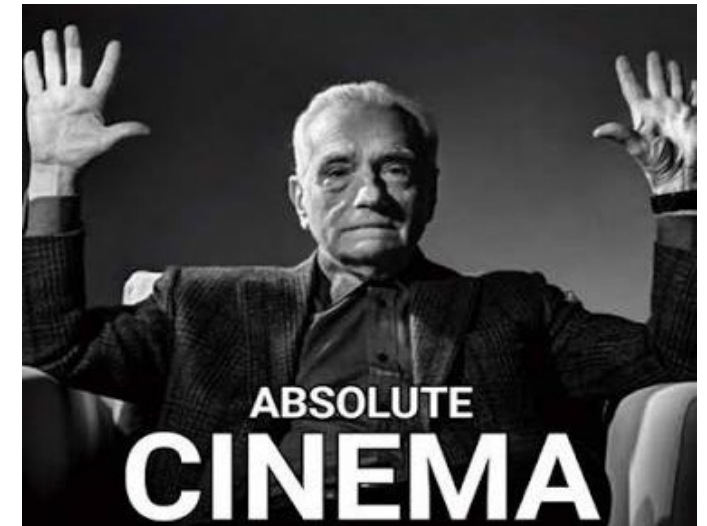
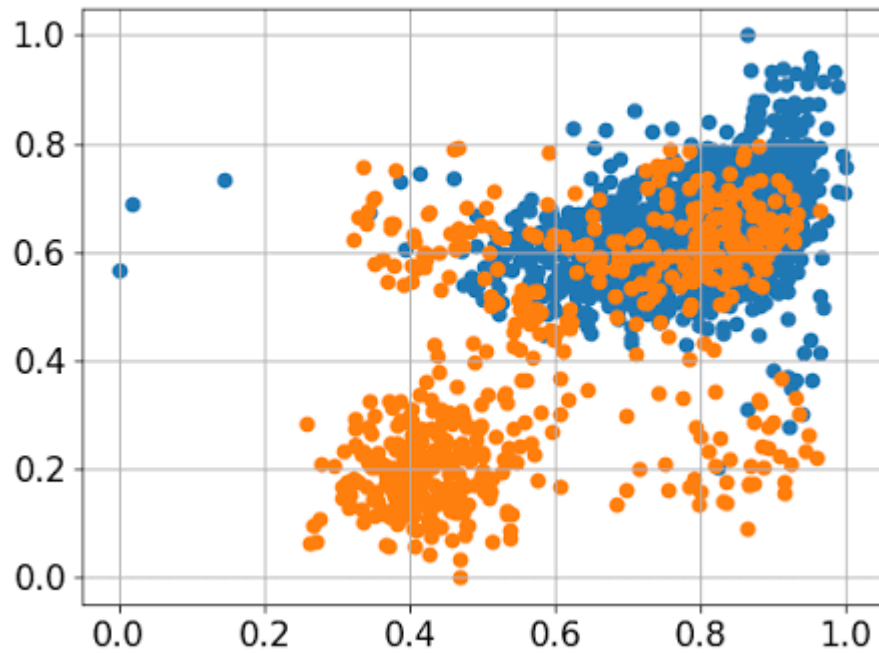
...and the same ramp photocopied — the order still survives



Okabe & Ito, Color Universal Design — jfly.uni-koeln.de/color/ | R: `palette.colors(palette = "Okabe-Ito")`

Image integrated by AI

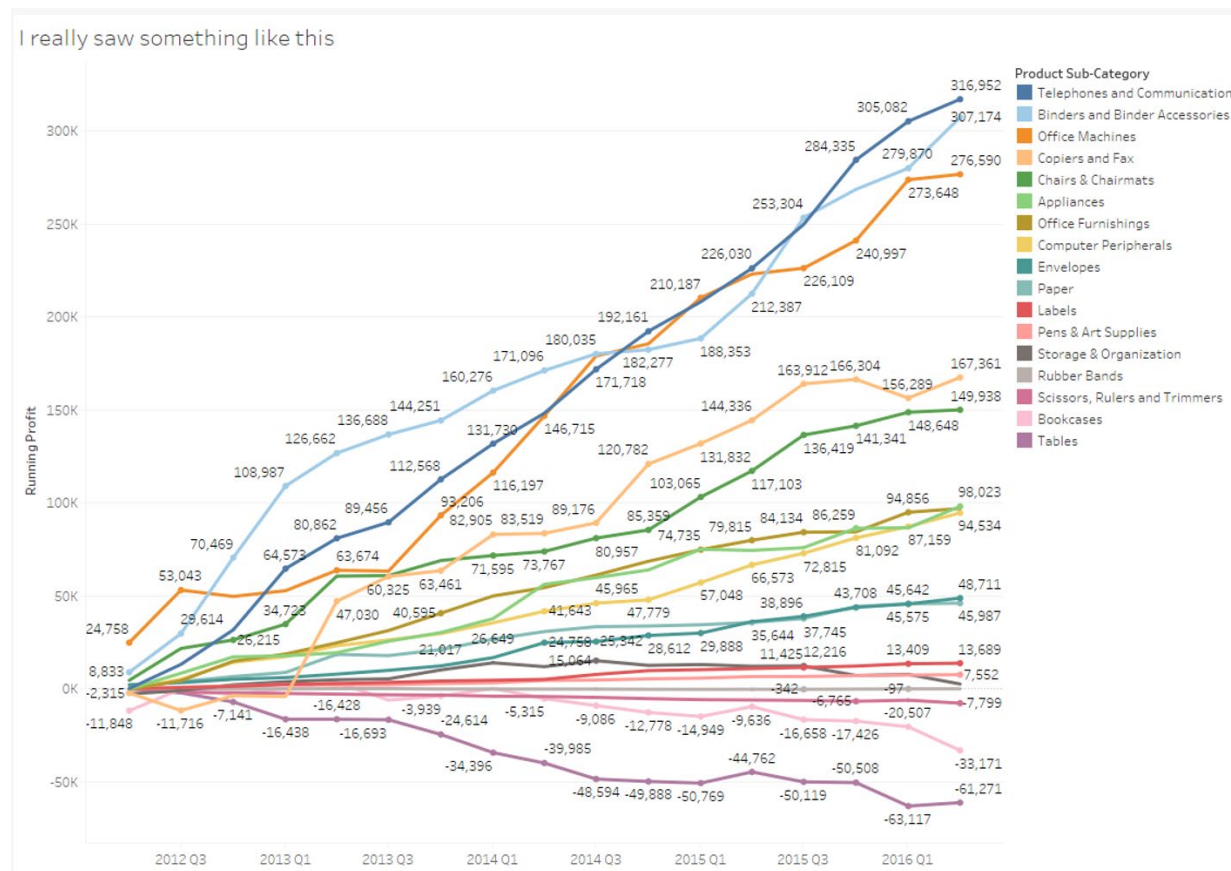
It's cinema!



Four considerations that cost you nothing

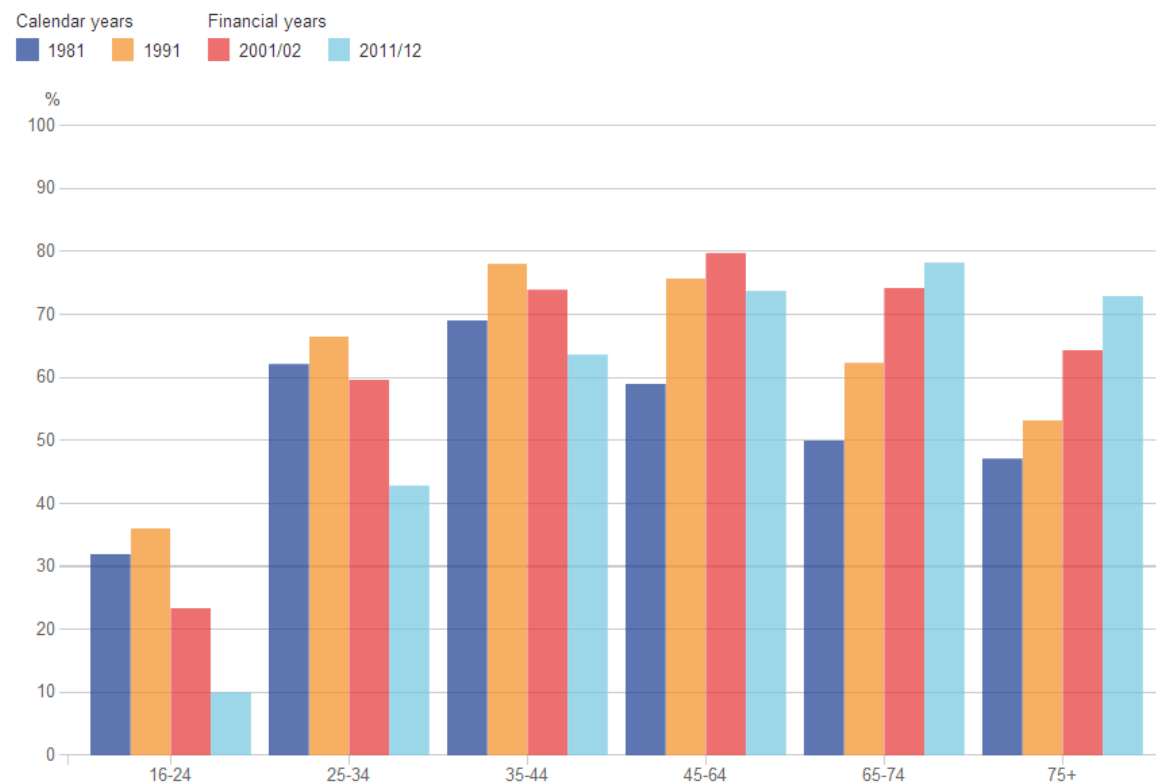
4. Use fewer colours

- At most four or five categories; nobody can hold the legend in their head (colour-blind or not).
- Colour the one or two series you are making a point about; grey out the rest.
- Depends on whether lines intersect, overlap or have sub-categories.

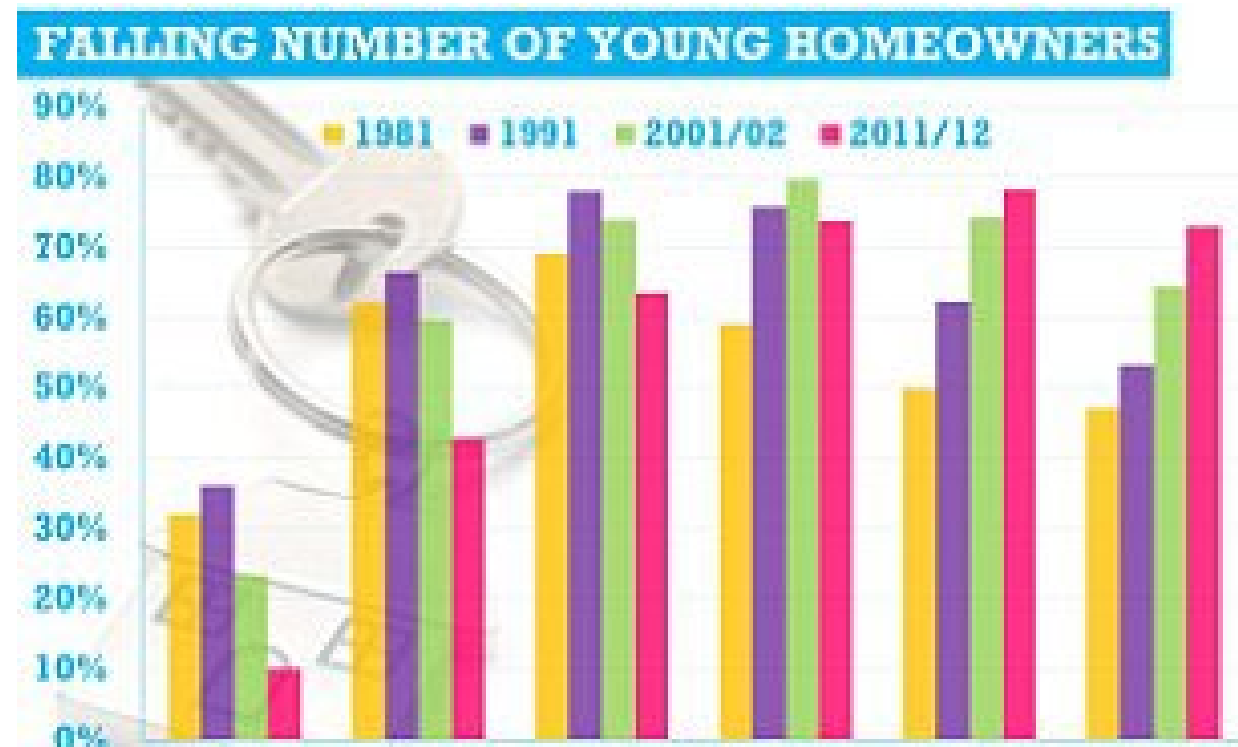


Back to this example

Percentage of each age group that are home owners⁵, England, 1981 to 2012



Source: English Housing Survey (EHS) 2012 to 2013, Table FC2101, DCLG ; EHS 2001/02, Table S106, DCLG



3. Elegant

- Is the design appealing?
 - *“Do not make something unless it is both necessary and useful; but if it is both, do not hesitate to make it beautiful!” (Frank Chimero, author)*
- Eliminate the arbitrary ≠ Make it minimalistic
- Decoration should be additive, not negative!

Smooth Sales

The online men's razor and blade market in the U.S. is growing, threatening traditional retail sales.

\$111M
2013

\$189M
2014

\$141M
2015, through May



Source: Slice Intelligence

Photo: F. Martin Ramin/The Wall Street Journal; styling by Anne Cardenas

THE WALL STREET JOURNAL.

- Innovative
- Long-lasting
- Environmentally friendly!

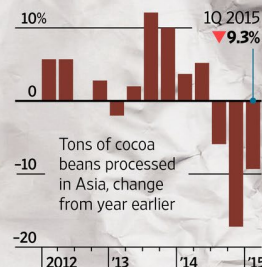
Dark Outlook

Rising chocolate prices in China...

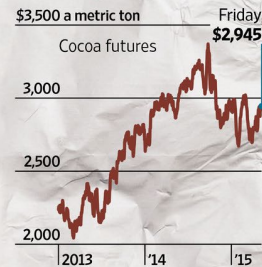
Retail prices for chocolate confections, per kilogram



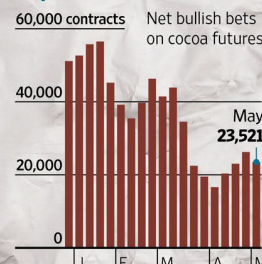
...are weighing on cocoa demand...



...pushing down prices...



...and prompting investors to pare back on bullish bets.



Sources: Euromonitor (retail prices); Cocoa Association of Asia (beans processed); FactSet (futures); U.S. Commodity Futures Trading Commission (bets)

THE WALL STREET JOURNAL.

Highest Ocean Plastic Waste Polluters

(annual estimation in metric ton)



Marine Tales

The Hidden Thinking

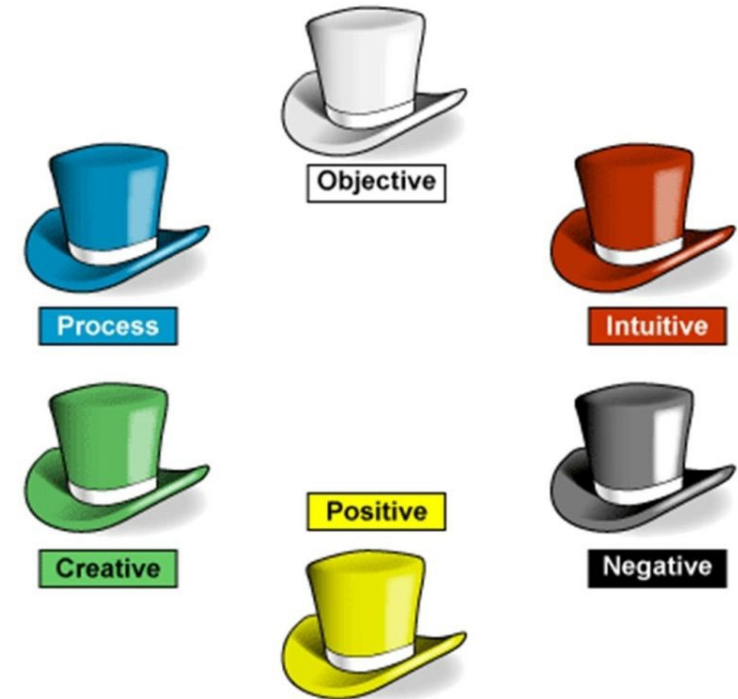
Step 1: Formulating your brief

What is your motivational curiosity?

- Some original interest held by someone about a subject
- You don't (only) create a viz because you happen to have data, but also because there is a desire for you to understand something!
- Do you have pet/passion projects?
- About what do you want to **facilitate understanding?**

Identifying Project Circumstances

- Frictions and freedoms that are imposed on you or determined by you
- Think about...
 - Stakeholders
 - Audience
 - Visualisers



The seven visualiser thinking hats

DIRECTOR | The coordinator, overseeing the project
 Initiates and leads on gathering and understanding requirements
 Identifies and establishes the project's key circumstances
 Defines the purpose of the project based on desired outcome
 Manages progress through the process and keeps it cohesive
 The primary decision maker, often needing to compromise
 Pays strong attention to detail
 Gets things done: checks, tests, finishes tasks

COMMUNICATOR | The broker between all people
 Helps to define the perspective of the audience
 A good listener with the humility to defer to domain experts
 Has a 'thick skin': needs patience, empathy and diplomacy
 A confident communicator with laypeople and non-specialists
 Possesses strong copy-editing abilities
 Manages expectations and presents possibilities
 Launches and promotes the final solution

JOURNALIST | The reporter, pursuing the scent of enquiry
 Driven by a desire to help others understand
 Defines the origin curiosity of the project
 Has an instinct to research, learn and discover
 Possesses or is able to acquire salient domain knowledge
 Understands the essence of the subject's data
 Has empathy for the interests and needs of an audience
 Defines the editorial angle, framing and focus

DATA ANALYST | The wrangler, handling the data work
 Has strong data and statistical literacy
 Possesses technical skills to acquire data from multiple sources
 Examines the physical properties of the data
 Undertakes initial descriptive analysis
 Transforms and prepares the data for its purpose
 Undertakes exploratory data analysis
 Has database and data modelling experience

DESIGNER | The conceiver, providing creative direction
 Establishes the initial creative pathway through defining purpose
 Harnesses initial mental visualisations: ideas and inspiration
 Has strong creative, graphic and illustration skills
 Understands the principles of user interface design
 Is fluent with the full array of possible design options
 Unifies the decision making across the design anatomy
 Has a relentless creative drive to keep innovating

SCIENTIST | The thinker, providing scientific rigour
 Brings a strong research mindset to the process
 Understands the science of visual perception
 Understands visualisation, statistical and data ethics
 Understands the influence of human factors
 Verifies/validates the integrity of all data and design decisions
 Demonstrates a systems thinking approach to problem solving
 Undertakes reflective evaluation and critique

TECHNOLOGIST | The developer, constructing the solution
 Possesses a repertoire of software and programming capabilities
 Has an appetite to acquire new technical solutions
 Possesses strong mathematical knowledge
 Can automate otherwise manually intensive processes
 Has the discipline to avoid feature creep
 Works on the prototyping and development of the solution
 Undertakes pre and post-launch testing, evaluation and support

Figure 3.3 The Attributes that Comprise the 'Seven Hats of Visualisation Design'

Or eight!

The 8 Hats of Data Visualisation Design



Constraints

- Timescales
- Pressures
- Design
- Technological

Deliverables

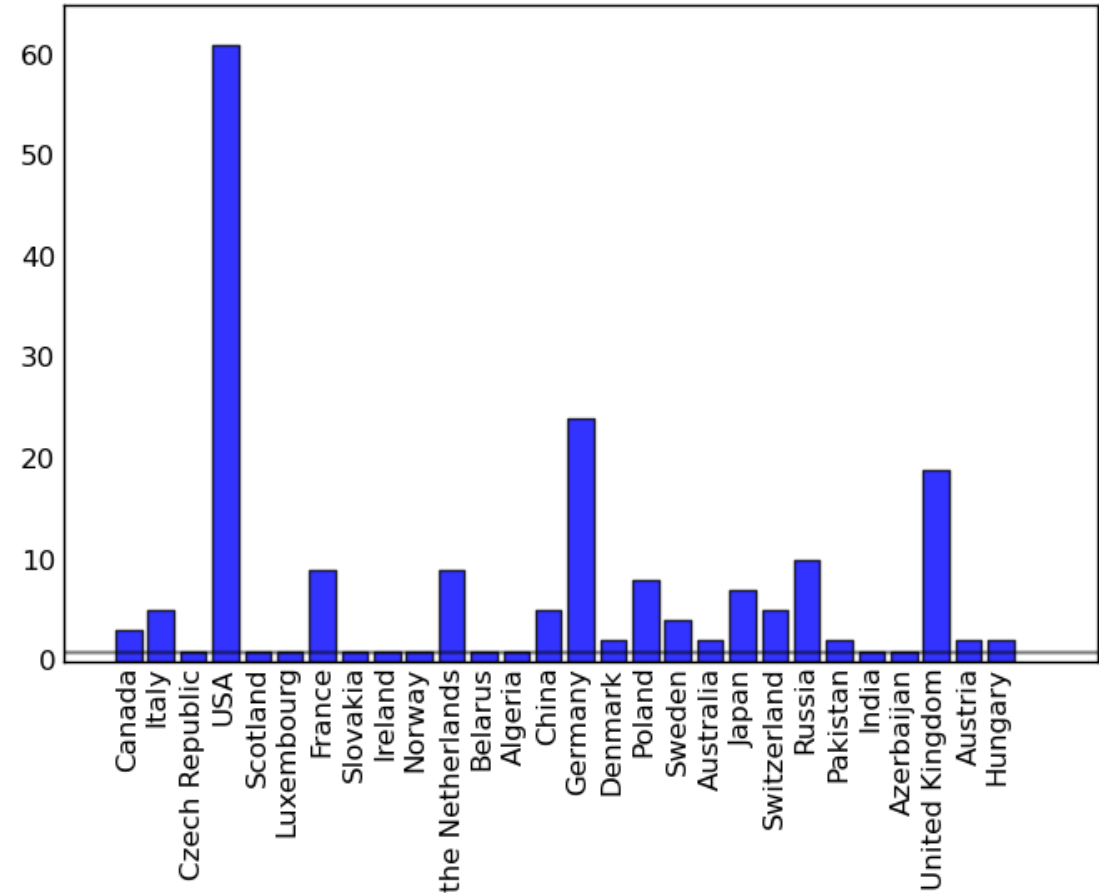
- Setting
- Medium
- Quantity
- Frequency

Defining your project's vision

- Mission $\neq$ Vision
- Two significant design characteristics
 - Tone: **Read or Feel**
 - Experience: **Explain, Exhibit or Explore**

Reading Tone

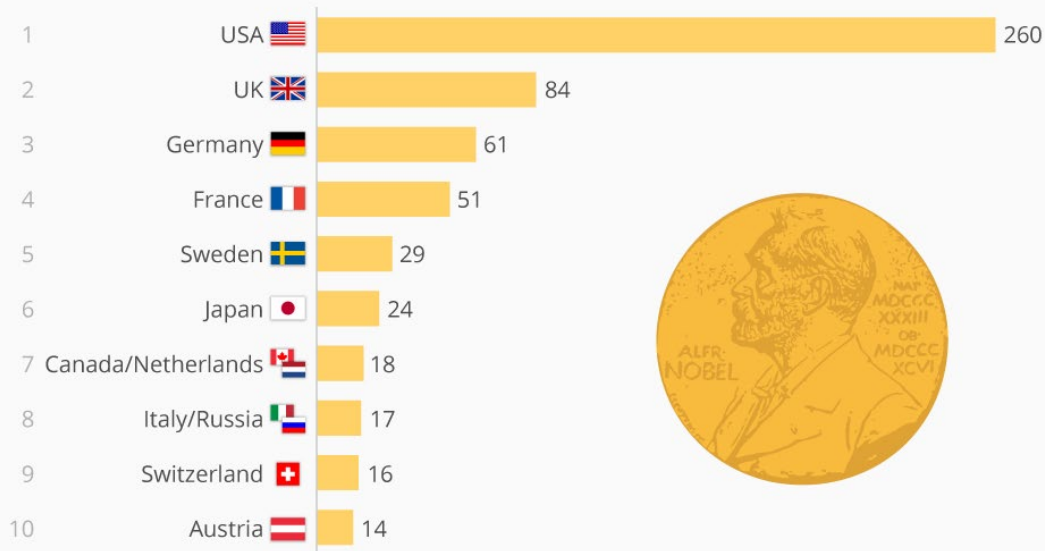
- Optimise the precision and efficiency to perceive the represented data
- Pragmatic, analytical, conservative, utilitarian, boring(?)
- Bar charts are the reading tone choice by excellence!



Moving slightly to the “feel” area

USA leads the way in the Nobel Prize

The Top 10 Nobel Prize Winners by Country of Birth

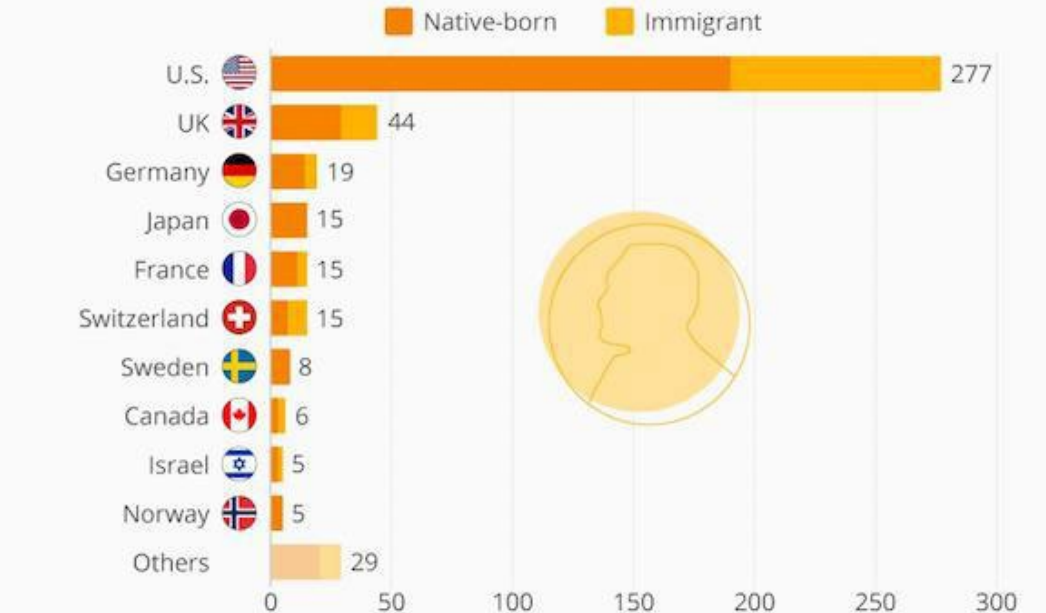


@StatistaCharts Source: Nobel Foundation

statista

Immigrants' Share of Nobel Prizes in the Sciences

Number of Nobel laureates (physics, chemistry, medicine, economy) per country 1969-2019*



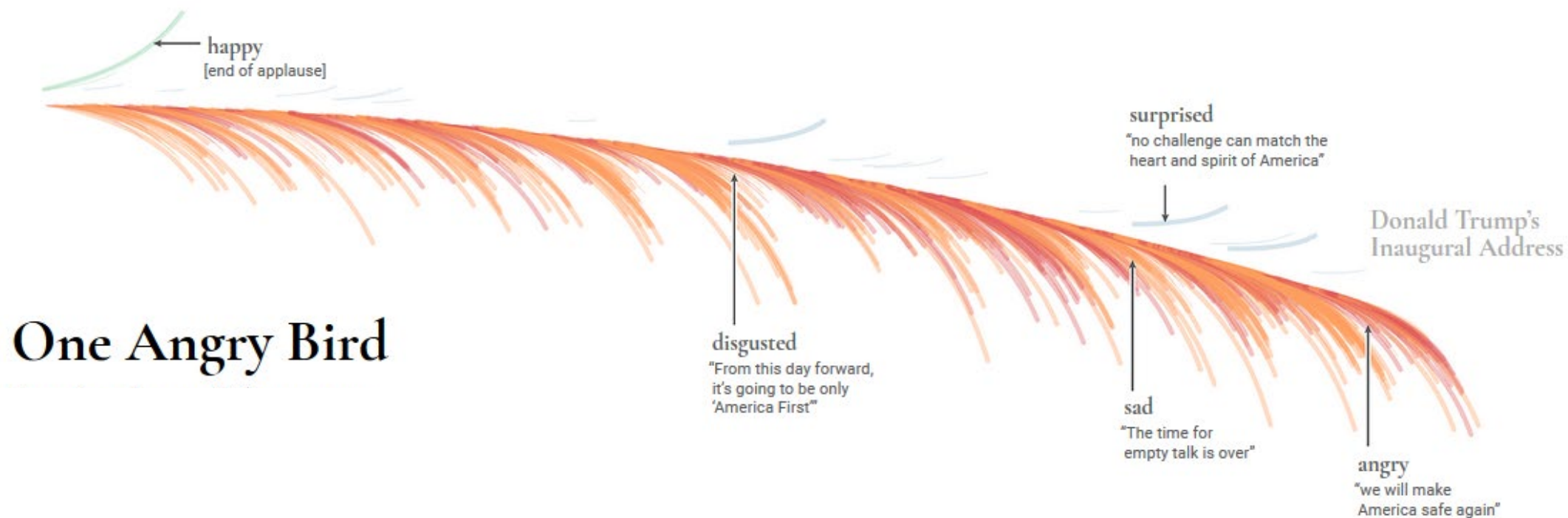
@StatistaCharts Source: Nobel Prize Foundation

* Immigration status determined by place of birth and affiliation at time of win

statista

Feeling Tone

- Emotive, figurative, seductive, big-picture, fun and dramatic
- Needs more in-depth analysis
- But that doesn't mean that you can't have a simple glance!



Taylor Swift is
quite often
and occ

Picture To Burn

You Belong With
Me

Mine

Dear

Spe

I Knew You Were
Trouble

2012
Red

Stay Stay Stay

Starlight

Shake It Off

Bad Blood

Wonderland

Delicate

Gorgeous

This Is Why We
Can't Have Nice
Things

New Year's Day

2025
The Life
of a
Showgirl



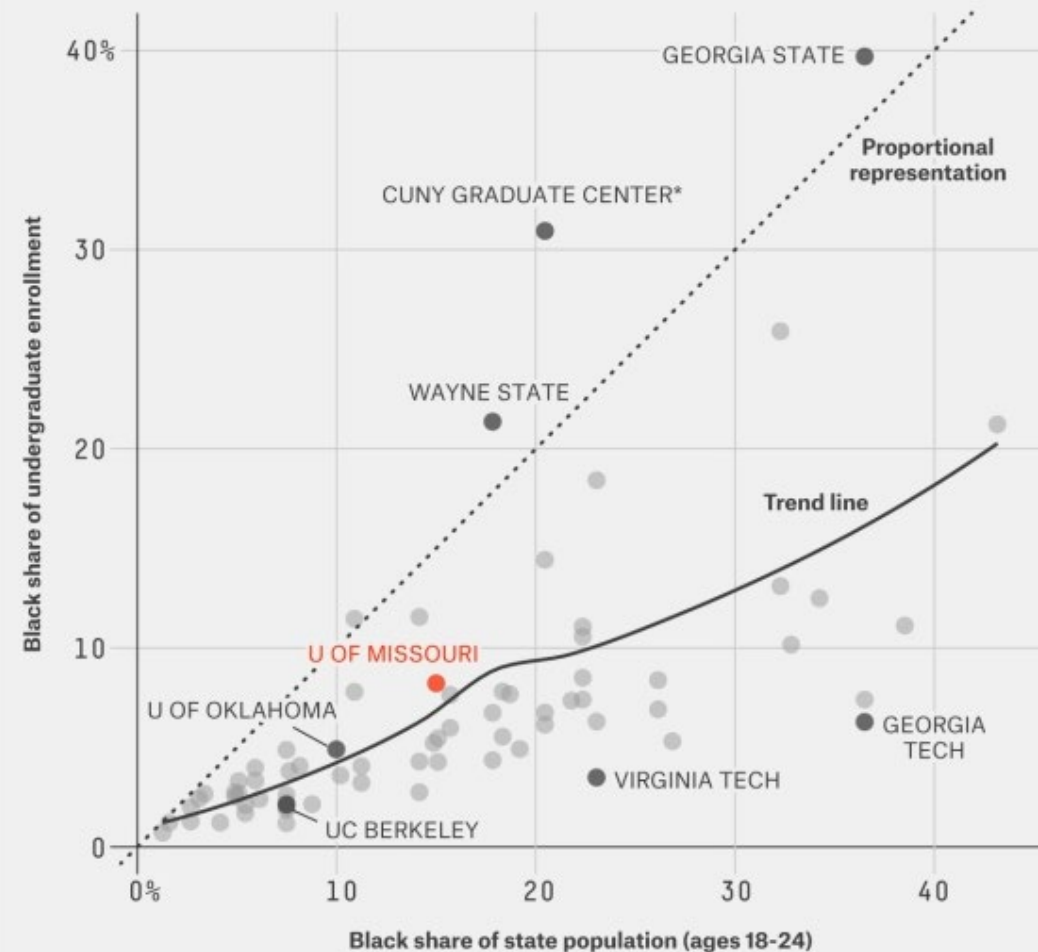
Experience

- Different methods of enabling interpretation
 - Explanatory
 - Exploratory
 - Exhibitory

Explanatory

Black Students Are Underrepresented On Campus

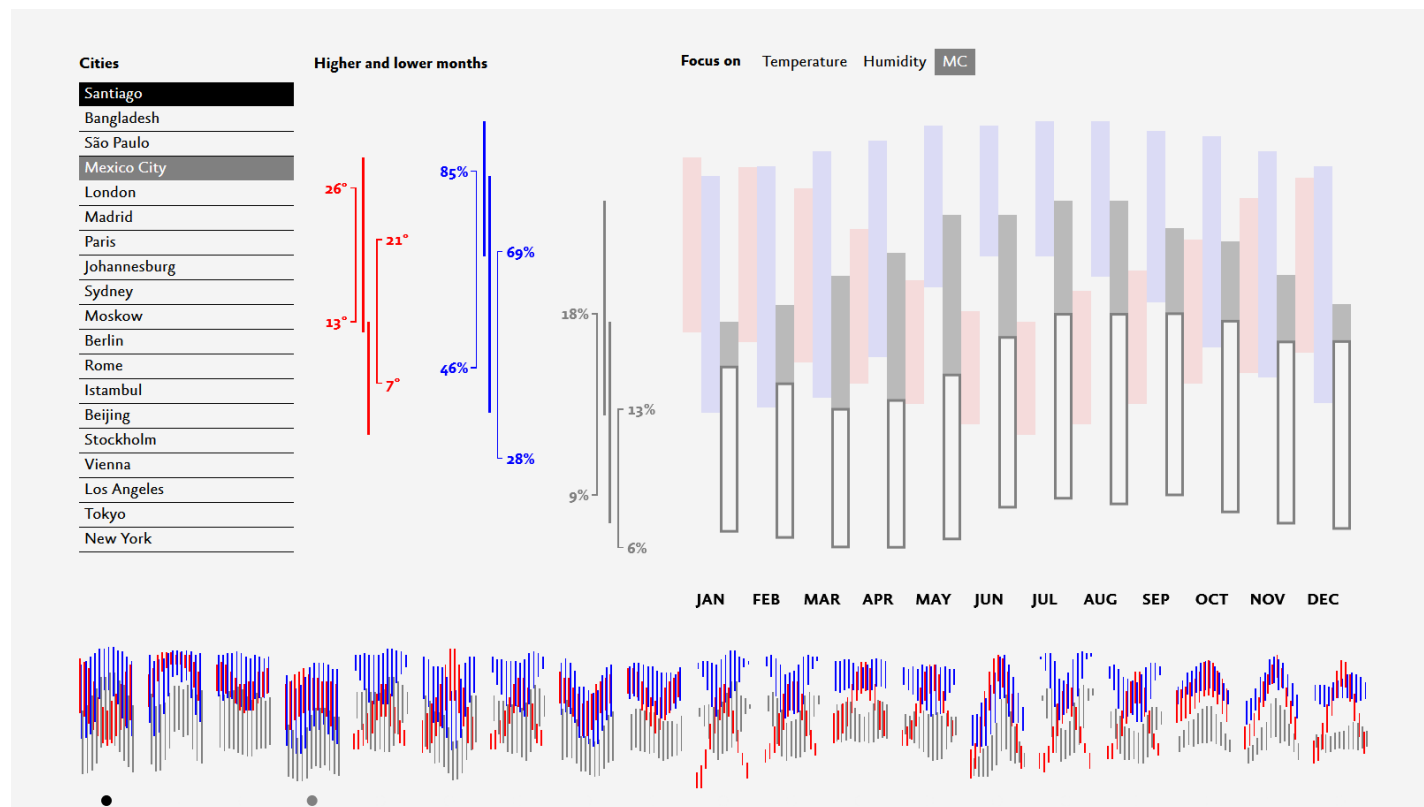
Black enrollment at public research universities vs. black college-age state population, 2013



*The CUNY Graduate Center primarily grants doctorates but has a small undergraduate population.

Exploratory

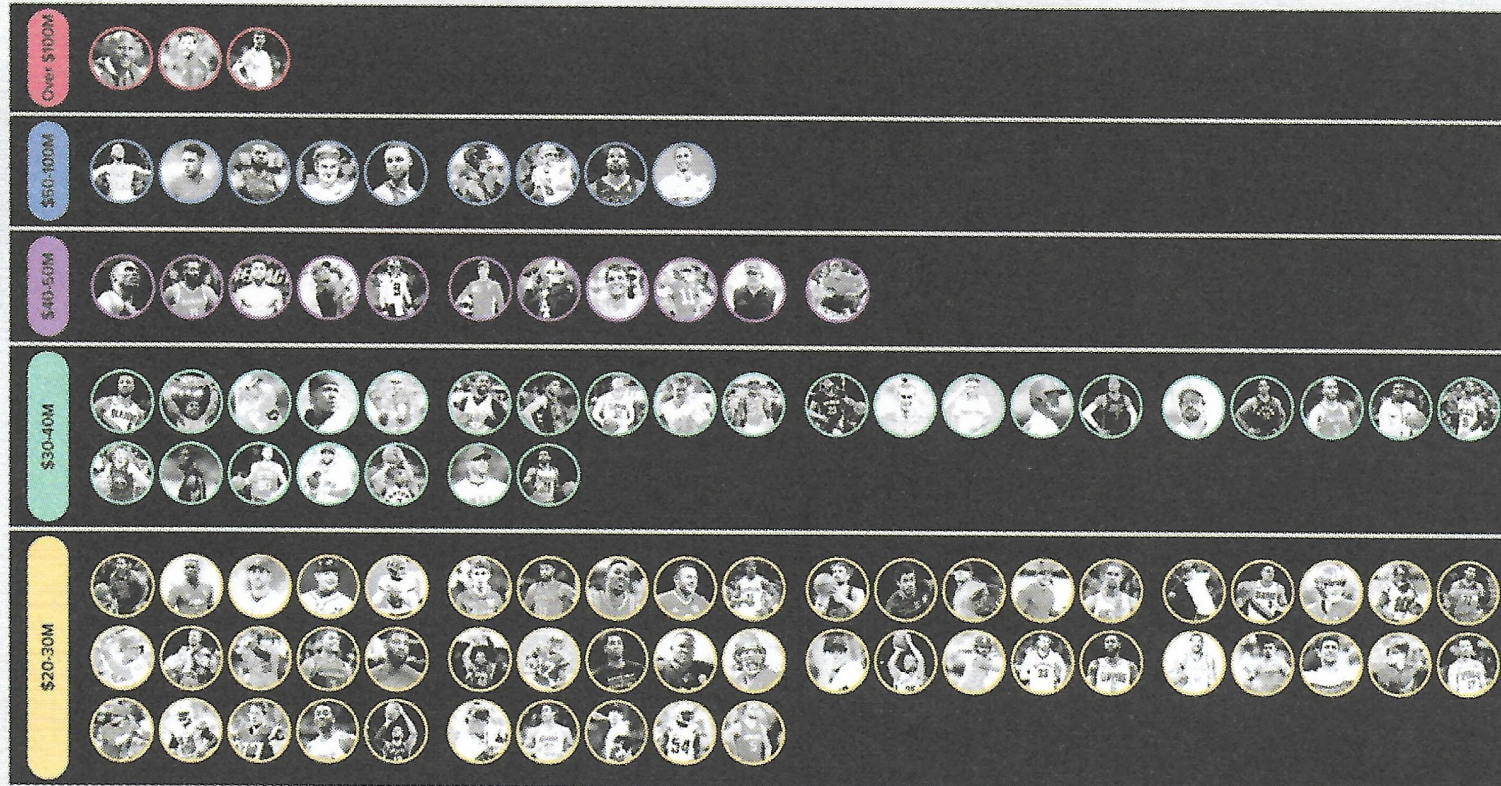
- Focused on helping the viewers discover and form their own interpretations
- Simple interpretation and manipulation of data
- A “good” example?
 - <https://luiscarli.com/2012/09/01/wood-changes/>



Exhibitory

Forbes: The World's 100 Highest-Paid Athletes

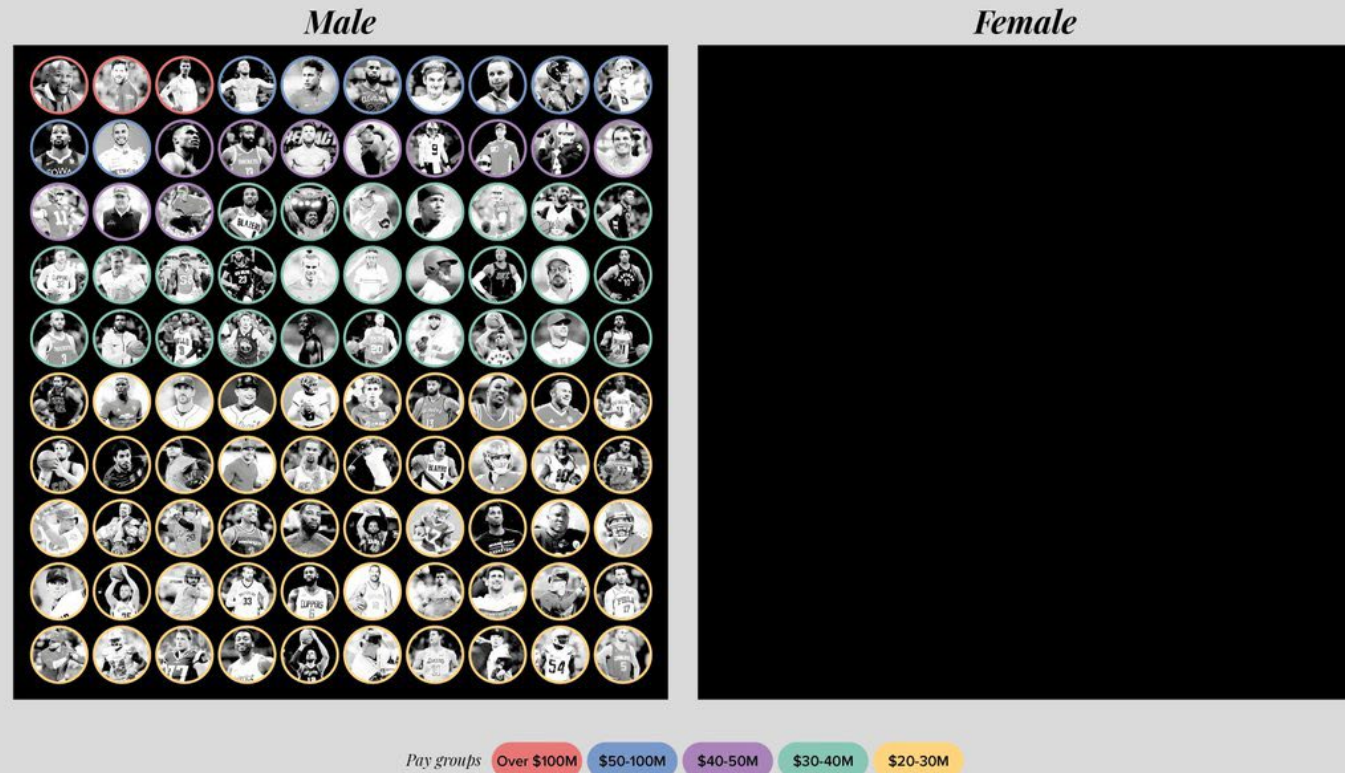
Pay groups



Data and images from <https://www.forbes.com/athletes/list/>

(even more) Exhibitory

Forbes: The World's 100 Highest-Paid Athletes



Data and images from <https://www.forbes.com/athletes/list/>

Another example:

<https://informationisbeautiful.net/visualizations/the-billion-dollar-o-gram-2009/>

Conclusions

Coursework Clarification

Lab
